

Name: Key

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Please circle your instructor's name: Kevin Meek James Gossell Margaret Short

There are 25 points possible on this quiz. No aids (book, calculator, etc.) are permitted. **Show all work for full credit.**

1. [15 points] Find the derivative of each function. You do not need to simplify your answer.

a. $f(\theta) = \arcsin(5\theta)$

$$f'(\theta) = \frac{1}{\sqrt{1-(5\theta)^2}} \cdot 5$$

b. $g(x) = \ln\left(\frac{x^5}{\sin(x)}\right) = \ln(x^5) - \ln(\sin x) = 5\ln x - \ln(\sin x)$

$$g'(x) = \frac{5}{x} - \frac{1}{\sin x} \cdot \cos x$$

c. $h(t) = \arctan(3t + t^2)$

$$h'(t) = \frac{1}{1 + (3t + t^2)^2} \cdot (3 + 2t)$$

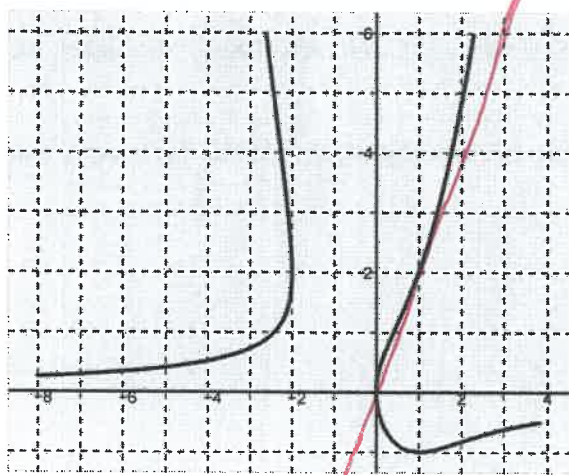
d. $f(x) = (\cos(x))^{-1} + \cos^{-1}(x) = \frac{1}{\cos x} + \arccos x = \sec x + \arccos x$

$$f'(x) = \sec x \tan x + \frac{-1}{\sqrt{1-x^2}}$$

e. $g(x) = e^{\ln(x)} + 5e^{3x+1} - 2^x + 6e^{21}$

$$g'(x) = e^{\ln x} \cdot \frac{1}{x} + 5e^{3x+1} \cdot (3) - 2^x \ln 2 + 0$$

2. [10 points] The graph of the curve $yx^2 = y^2 - 2x$ appears below.



- a. Use implicit differentiation to find $\frac{dy}{dx}$.

$$\frac{d}{dx}[yx^2] = \frac{d}{dx}[y^2 - 2x]$$

$$\frac{dy}{dx} \cdot x^2 + y(2x) = 2y \cdot \frac{dy}{dx} - 2$$

$$2xy + 2 = 2y \cdot \frac{dy}{dx} - x^2 \cdot \frac{dy}{dx}$$

$$2xy + 2 = \frac{dy}{dx}(2y - x^2)$$

$$\frac{dy}{dx} = \frac{2xy + 2}{2y - x^2}$$

- b. Find the equation of the tangent line to the curve at $P(1,2)$.

$$m = \left. \frac{dy}{dx} \right|_{(1,2)} = \frac{2(1)(2) + 2}{2(2) - (1)^2} = \frac{6}{3} = 2$$

$$y - 2 = 2(x - 1)$$

$$y = 2x$$

- c. Sketch the tangent line to at curve at $P(1,2)$ on the graph above.

3. [2 points] BONUS: Let $f(x) = x^{3x}$, $x > 0$. Find $f'(x)$. Let $y = f(x)$.

$$y = x^{3x}$$

$$\ln y = \ln x^{3x}$$

$$\ln y = 3x \ln x$$

$$\frac{d}{dx}[\ln y] = \frac{d}{dx}[3x \ln x]$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = 3 \ln x + 3x \cdot \frac{1}{x}$$

$$\frac{dy}{dx} = y(3 \ln x + 3)$$

$$\frac{dy}{dx} = x^{3x}(3 \ln x + 3)$$

$$f'(x) = x^{3x}(3 \ln x + 3)$$

